

A guide to the internals of the GNU linker

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This file documents the internals of the GNU linker `ld`. It is a collection of miscellaneous information with little form at this point. Mostly, it is a repository into which you can put information about GNU `ld` as you discover it (or as you design changes to `ld`).

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1 The ‘README’ File

Check the ‘README’ file; it often has useful information that does not appear anywhere else in the directory.

2 How linker emulations are generated

Each linker target has an *emulation*. The emulation includes the default linker script, and certain emulations also modify certain types of linker behaviour.

Emulations are created during the build process by the shell script ‘genscripts.sh’.

The ‘genscripts.sh’ script starts by reading a file in the ‘emulparams’ directory. This is a shell script which sets various shell variables used by ‘genscripts.sh’ and the other shell scripts it invokes.

The ‘genscripts.sh’ script will invoke a shell script in the ‘scripttempl’ directory in order to create default linker scripts written in the linker command language. The ‘scripttempl’ script will be invoked 5 (or, in some cases, 6) times, with different assignments to shell variables, to create different default scripts. The choice of script is made based on the command-line options.

After creating the scripts, ‘genscripts.sh’ will invoke yet another shell script, this time in the ‘emultempl’ directory. That shell script will create the emulation source file, which contains C code. This C code permits the linker emulation to override various linker behaviours. Most targets use the generic emulation code, which is in ‘emultempl/generic.em’.

To summarize, ‘genscripts.sh’ reads three shell scripts: an emulation parameters script in the ‘emulparams’ directory, a linker script generation script in the ‘scripttempl’ directory, and an emulation source file generation script in the ‘emultempl’ directory.

For example, the Sun 4 linker sets up variables in ‘emulparams/sun4.sh’, creates linker scripts using ‘scripttempl/aout.sc’, and creates the emulation code using ‘emultempl/sunos.em’.

Note that the linker can support several emulations simultaneously, depending upon how it is configured. An emulation can be selected with the `-m` option. The `-V` option will list all supported emulations.

2.1 ‘emulparams’ scripts

Each target selects a particular file in the ‘emulparams’ directory by setting the shell variable `targ_emul` in ‘configure.tgt’. This shell variable is used by the ‘configure’ script to control building an emulation source file.

Certain conventions are enforced. Suppose the `targ_emul` variable is set to *emul* in ‘configure.tgt’. The name of the emulation shell script will be ‘emulparams/emul.sh’. The ‘Makefile’ must have a target named ‘eemul.c’; this target must depend upon ‘emulparams/emul.sh’, as well as the appropriate scripts in the ‘scripttempl’ and ‘emultempl’ directories. The ‘Makefile’ target must invoke `GENSCRIPTS` with two arguments: *emul*, and the value of the make variable `tdir_emul`. The value of the latter variable will be set by the ‘configure’ script, and is used to set the default target directory to search.

By convention, the ‘emulparams/emul.sh’ shell script should only set shell variables. It may set shell variables which are to be interpreted by the ‘scripttempl’ and the ‘emultempl’ scripts. Certain shell variables are interpreted directly by the ‘genscripts.sh’ script.

Here is a list of shell variables interpreted by ‘genscripts.sh’, as well as some conventional shell variables interpreted by the ‘scripttempl’ and ‘emultempl’ scripts.

SCRIPT_NAME

This is the name of the ‘scripttempl’ script to use. If `SCRIPT_NAME` is set to *script*, ‘genscripts.sh’ will use the script ‘scripttempl/script.sc’.

TEMPLATE_NAME

This is the name of the ‘emultempl’ script to use. If `TEMPLATE_NAME` is set to *template*, ‘genscripts.sh’ will use the script ‘emultempl/template.em’. If this variable is not set, the default value is ‘generic’.

GENERATE_SHLIB_SCRIPT

If this is set to a nonempty string, ‘genscripts.sh’ will invoke the ‘scripttempl’ script an extra time to create a shared library script.
Section 2.2 [linker scripts], page 3.

OUTPUT_FORMAT

This is normally set to indicate the BFD output format use (e.g., “a.out-sunos-big”). The ‘scripttempl’ script will normally use it in an `OUTPUT_FORMAT` expression in the linker script.

ARCH

This is normally set to indicate the architecture to use (e.g., ‘sparc’). The ‘scripttempl’ script will normally use it in an `OUTPUT_ARCH` expression in the linker script.

ENTRY

Some ‘scripttempl’ scripts use this to set the entry address, in an `ENTRY` expression in the linker script.

TEXT_START_ADDR

Some ‘scripttempl’ scripts use this to set the start address of the ‘.text’ section.

SEGMENT_SIZE

The ‘genscripts.sh’ script uses this to set the default value of `DATA_ALIGNMENT` when running the ‘scripttempl’ script.

TARGET_PAGE_SIZE

If **SEGMENT_SIZE** is not defined, the `'genscripts.sh'` script uses this to define it.

ALIGNMENT

Some `'scripttempl'` scripts set this to a number to pass to **ALIGN** to set the required alignment for the **end** symbol.

2.2 `'scripttempl'` scripts

Each linker target uses a `'scripttempl'` script to generate the default linker scripts. The name of the `'scripttempl'` script is set by the **SCRIPT_NAME** variable in the `'emulparams'` script. If **SCRIPT_NAME** is set to *script*, `genscripts.sh` will invoke `'scripttempl/script.sc'`.

The `'genscripts.sh'` script will invoke the `'scripttempl'` script 5 to 9 times. Each time it will set the shell variable **LD_FLAG** to a different value. When the linker is run, the options used will direct it to select a particular script. (Script selection is controlled by the `get_script` emulation entry point; this describes the conventional behaviour).

The `'scripttempl'` script should just write a linker script, written in the linker command language, to standard output. If the emulation name—the name of the `'emulparams'` file without the `'.sc'` extension—is *emul*, then the output will be directed to `'ldscripts/emul.extension'` in the build directory, where *extension* changes each time the `'scripttempl'` script is invoked.

Here is the list of values assigned to **LD_FLAG**.

- | | |
|---------------|---|
| (empty) | The script generated is used by default (when none of the following cases apply). The output has an extension of <code>'.x'</code> . |
| n | The script generated is used when the linker is invoked with the <code>-n</code> option. The output has an extension of <code>'.xn'</code> . |
| N | The script generated is used when the linker is invoked with the <code>-N</code> option. The output has an extension of <code>'.xN'</code> . |
| r | The script generated is used when the linker is invoked with the <code>-r</code> option. The output has an extension of <code>'.xr'</code> . |
| u | The script generated is used when the linker is invoked with the <code>-Ur</code> option. The output has an extension of <code>'.xu'</code> . |
| shared | The <code>'scripttempl'</code> script is only invoked with LD_FLAG set to this value if GENERATE_SHLIB_SCRIPT is defined in the <code>'emulparams'</code> file. The <code>'emultempl'</code> script must arrange to use this script at the appropriate time, normally when the linker is invoked with the <code>-shared</code> option. The output has an extension of <code>'.xs'</code> . |
| c | The <code>'scripttempl'</code> script is only invoked with LD_FLAG set to this value if GENERATE_COMBRELOC_SCRIPT is defined in the <code>'emulparams'</code> file or if SCRIPT_NAME is <code>elf</code> . The <code>'emultempl'</code> script must arrange to use this script at the appropriate time, normally when the linker is invoked with the <code>-z combreloc</code> option. The output has an extension of <code>'.xc'</code> . |

- cshared** The ‘scripttempl’ script is only invoked with LD_FLAG set to this value if GENERATE_COMBRELOC_SCRIPT is defined in the ‘emulparams’ file or if SCRIPT_NAME is elf and GENERATE_SHLIB_SCRIPT is defined in the ‘emulparams’ file. The ‘emultempl’ script must arrange to use this script at the appropriate time, normally when the linker is invoked with the `-shared -z combreloc` option. The output has an extension of ‘.xsc’.
- wshared** Like *cshared* but in addition to needing the ‘-shared’ and ‘-z combreloc’ options, it also needs the ‘-z relro’ option. If the ‘-z now’ option is also used then the output has an extension of ‘.xsw’, otherwise it is ‘.xsco’.
- pie** The ‘scripttempl’ script is only invoked with LD_FLAG set to this value if GENERATE_PIE_SCRIPT is defined in the ‘emulparams’ file. The ‘emultempl’ script must arrange to use this script at the appropriate time, normally when the linker is invoked with the `-pie` option. The output has an extension of ‘.xd’.
- In addition if the ‘scripttempl’ script is also invoked with GENERATE_RELRO_SCRIPT defined then a second script is created with an extension of ‘.xdo’ to match the ‘-pie’ and ‘-z relro’ options.
- cpie** Like *pie* but this version is used when the ‘scripttempl’ script is invoked with both GENERATE_PIE_SCRIPT and GENERATE_COMBRELOC_SCRIPT defined. The ‘emultempl’ script must arrange to use this script at the appropriate time, normally when the linker is invoked with the `-pie` and ‘-z combreloc’ options. The output has an extension of ‘.xdc’.
- wpie** Like *cpie*, but in addition the ‘-z relro’ and ‘-z now’ options need to be defined. The output has an extension of ‘.xdw’.

textonly

pietextonly

cpietextonly

wpietextonly

ctextonly

wtextonly

sharedtextonly

csharedtextonly

wsharedtextonly

rotextonly

roctextonly

rowtextonly

rosharedtextonly

rocsharedtextonly

rowsharedtextonly

ropietextonly

rocpietextonly

rowpietextonly

auto_import

The ‘scripttempl’ script is only invoked with LD_FLAG set to this value if GENERATE_AUTO_IMPORT_SCRIPT is defined in the ‘emulparams’ file. The

‘**emultempl**’ script must arrange to use this script at the appropriate time, normally when the linker is invoked with the **--enable-auto-import** option. The output has an extension of ‘**.xa**’.

Besides the shell variables set by the ‘**emulparams**’ script, and the **LD_FLAG** variable, the ‘**genscripts.sh**’ script will set certain variables for each run of the ‘**scripttempl**’ script.

RELOCATING

This will be set to a non-empty string when the linker is doing a final relocation (e.g., all scripts other than **-r** and **-Ur**).

CONSTRUCTING

This will be set to a non-empty string when the linker is building global constructor and destructor tables (e.g., all scripts other than **-r**).

DATA_ALIGNMENT

This will be set to an **ALIGN** expression when the output should be page aligned, or to ‘.’ when generating the **-N** script.

CREATE_SHLIB

This will be set to a non-empty string when generating a **-shared** script.

COMBRELOC

This will be set to a non-empty string when generating **-z combreloc** scripts to a temporary file name which can be used during script generation.

The conventional way to write a ‘**scripttempl**’ script is to first set a few shell variables, and then write out a linker script using **cat** with a here document. The linker script will use variable substitutions, based on the above variables and those set in the ‘**emulparams**’ script, to control its behaviour.

When there are parts of the ‘**scripttempl**’ script which should only be run when doing a final relocation, they should be enclosed within a variable substitution based on **RELOCATING**. For example, on many targets special symbols such as **_end** should be defined when doing a final link. Naturally, those symbols should not be defined when doing a relocatable link using **-r**. The ‘**scripttempl**’ script could use a construct like this to define those symbols:

```
{RELOCATING+ _end = .;}
```

This will do the symbol assignment only if the **RELOCATING** variable is defined.

The basic job of the linker script is to put the sections in the correct order, and at the correct memory addresses. For some targets, the linker script may have to do some other operations.

For example, on most MIPS platforms, the linker is responsible for defining the special symbol **_gp**, used to initialize the **\$gp** register. It must be set to the start of the small data section plus **0x8000**. Naturally, it should only be defined when doing a final relocation. This will typically be done like this:

```
{RELOCATING+ _gp = ALIGN(16) + 0x8000;}
```

This line would appear just before the sections which compose the small data section (‘**.sdata**’, ‘**.sbss**’). All those sections would be contiguous in memory.

Many COFF systems build constructor tables in the linker script. The compiler will arrange to output the address of each global constructor in a ‘**.ctor**’ section, and the

address of each global destructor in a `‘.dtor’` section (this is done by defining `ASM_OUTPUT_CONSTRUCTOR` and `ASM_OUTPUT_DESTRUCTOR` in the `gcc` configuration files). The `gcc` runtime support routines expect the constructor table to be named `__CTOR_LIST__`. They expect it to be a list of words, with the first word being the count of the number of entries. There should be a trailing zero word. (Actually, the count may be -1 if the trailing word is present, and the trailing word may be omitted if the count is correct, but, as the `gcc` behaviour has changed slightly over the years, it is safest to provide both). Here is a typical way that might be handled in a `‘scripttempl’` file.

```
{CONSTRUCTING+ __CTOR_LIST__ = .;}
{CONSTRUCTING+ LONG((__CTOR_END__ - __CTOR_LIST__) / 4 - 2)}
{CONSTRUCTING+ *(.ctors)}
{CONSTRUCTING+ LONG(0)}
{CONSTRUCTING+ __CTOR_END__ = .;}
{CONSTRUCTING+ __DTOR_LIST__ = .;}
{CONSTRUCTING+ LONG((__DTOR_END__ - __DTOR_LIST__) / 4 - 2)}
{CONSTRUCTING+ *(.dtors)}
{CONSTRUCTING+ LONG(0)}
{CONSTRUCTING+ __DTOR_END__ = .;}
```

The use of `CONSTRUCTING` ensures that these linker script commands will only appear when the linker is supposed to be building the constructor and destructor tables. This example is written for a target which uses 4 byte pointers.

Embedded systems often need to set a stack address. This is normally best done by using the `PROVIDE` construct with a default stack address. This permits the user to easily override the stack address using the `--defsym` option. Here is an example:

```
{RELOCATING+ PROVIDE (__stack = 0x80000000);}
```

The value of the symbol `__stack` would then be used in the startup code to initialize the stack pointer.

2.3 ‘emultempl’ scripts

Each linker target uses an `‘emultempl’` script to generate the emulation code. The name of the `‘emultempl’` script is set by the `TEMPLATE_NAME` variable in the `‘emulparams’` script. If the `TEMPLATE_NAME` variable is not set, the default is `‘generic’`. If the value of `TEMPLATE_NAME` is `template`, `‘genscripts.sh’` will use `‘emultempl/template.em’`.

Most targets use the generic `‘emultempl’` script, `‘emultempl/generic.em’`. A different `‘emultempl’` script is only needed if the linker must support unusual actions, such as linking against shared libraries.

The `‘emultempl’` script is normally written as a simple invocation of `cat` with a here document. The document will use a few variable substitutions. Typically each function names uses a substitution involving `EMULATION_NAME`, for ease of debugging when the linker supports multiple emulations.

Every function and variable in the emitted file should be static. The only globally visible object must be named `ld_EMULATION_NAME_emulation`, where `EMULATION_NAME` is the name of the emulation set in `‘configure.tgt’` (this is also the name of the `‘emulparams’` file without the `‘.sh’` extension). The `‘genscripts.sh’` script will set the shell variable `EMULATION_NAME` before invoking the `‘emultempl’` script.

The `ld_EMULATION_NAME_emulation` variable must be a `struct ld_emulation_xfer_struct`, as defined in `‘ldemul.h’`. It defines a set of function pointers which are invoked by

the linker, as well as strings for the emulation name (normally set from the shell variable `EMULATION_NAME` and the default BFD target name (normally set from the shell variable `OUTPUT_FORMAT` which is normally set by the ‘`emulparams`’ file).

The ‘`genscripts.sh`’ script will set the shell variable `COMPILE_IN` when it invokes the ‘`emultempl`’ script for the default emulation. In this case, the ‘`emultempl`’ script should include the linker scripts directly, and return them from the `get_scripts` entry point. When the emulation is not the default, the `get_scripts` entry point should just return a file name. See ‘`emultempl/generic.em`’ for an example of how this is done.

At some point, the linker emulation entry points should be documented.

3 A Walkthrough of a Typical Emulation

This chapter is to help people who are new to the way emulations interact with the linker, or who are suddenly thrust into the position of having to work with existing emulations. It will discuss the files you need to be aware of. It will tell you when the given "hooks" in the emulation will be called. It will, hopefully, give you enough information about when and how things happen that you'll be able to get by. As always, the source is the definitive reference to this.

The starting point for the linker is in ‘`ldmain.c`’ where `main` is defined. The bulk of the code that's emulation specific will initially be in `emultempl/emulation.em` but will end up in `eemulation.c` when the build is done. Most of the work to select and interface with emulations is in `ldemul.h` and `ldemul.c`. Specifically, `ldemul.h` defines the `ld_emulation_xfer_struct` structure your emulation exports.

Your emulation file exports a symbol `ld_EMULATION_NAME_emulation`. If your emulation is selected (it usually is, since usually there's only one), `ldemul.c` sets the variable `ld_emulation` to point to it. `ldemul.c` also defines a number of API functions that interface to your emulation, like `ldemul_after_parse` which simply calls your `ld_EMULATION_NAME_emulation.after_parse` function. For the rest of this section, the functions will be mentioned, but you should assume the indirect reference to your emulation also.

We will also skip or gloss over parts of the link process that don't relate to emulations, like setting up internationalization.

After initialization, `main` selects an emulation by pre-scanning the command-line arguments. It calls `ldemul_choose_target` to choose a target. If you set `choose_target` to `ldemul_default_target`, it picks your `target_name` by default.

`main` calls `ldemul_before_parse`, then `parse_args`. `parse_args` calls `ldemul_parse_args` for each arg, which must update the `getopt` globals if it recognizes the argument. If the emulation doesn't recognize it, then `parse_args` checks to see if it recognizes it.

Now that the emulation has had access to all its command-line options, `main` calls `ldemul_set_symbols`. This can be used for any initialization that may be affected by options. It is also supposed to set up any variables needed by the emulation script.

`main` now calls `ldemul_get_script` to get the emulation script to use (based on arguments, no doubt, see [Chapter 2 \[Emulations\]](#), page 1) and runs it. While parsing, `ldgram.y` may call `ldemul_hll` or `ldemul_syslib` to handle the HLL or SYSLIB commands. It may call `ldemul_unrecognized_file` if you asked the linker to link a file it doesn't recognize.

It will call `ldemul_recognized_file` for each file it does recognize, in case the emulation wants to handle some files specially. All the while, it's loading the files (possibly calling `ldemul_open_dynamic_archive`) and symbols and stuff. After it's done reading the script, `main` calls `ldemul_after_parse`. Use the after-parse hook to set up anything that depends on stuff the script might have set up, like the entry point.

`main` next calls `lang_process` in `ldlang.c`. This appears to be the main core of the linking itself, as far as emulation hooks are concerned(*). It first opens the output file's BFD, calling `ldemul_set_output_arch`, and calls `ldemul_create_output_section_statements` in case you need to use other means to find or create object files (i.e. shared libraries found on a path, or fake stub objects). Despite the name, nobody creates output sections here.

(*) In most cases, the BFD library does the bulk of the actual linking, handling symbol tables, symbol resolution, relocations, and building the final output file. See the BFD reference for all the details. Your emulation is usually concerned more with managing things at the file and section level, like "put this here, add this section", etc.

Next, the objects to be linked are opened and BFDs created for them, and `ldemul_after_open` is called. At this point, you have all the objects and symbols loaded, but none of the data has been placed yet.

Next comes the Big Linking Thingy (except for the parts BFD does). All input sections are mapped to output sections according to the script. If a section doesn't get mapped by default, `ldemul_place_orphan` will get called to figure out where it goes. Next it figures out the offsets for each section, calling `ldemul_before_allocation` before and `ldemul_after_allocation` after deciding where each input section ends up in the output sections.

The last part of `lang_process` is to figure out all the symbols' values. After assigning final values to the symbols, `ldemul_finish` is called, and after that, any undefined symbols are turned into fatal errors.

OK, back to `main`, which calls `ldwrite` in `ldwrite.c`. `ldwrite` calls BFD's `final_link`, which does all the relocation fixups and writes the output bfd to disk, and we're done.

In summary,

- `main()` in `ldmain.c`
- `'emultempl/EMULATION.em'` has your code
- `ldemul_choose_target` (defaults to your `target_name`)
- `ldemul_before_parse`
- Parse argv, calls `ldemul_parse_args` for each
- `ldemul_set_symbols`
- `ldemul_get_script`
- parse script
 - may call `ldemul_hll` or `ldemul_syslib`
 - may call `ldemul_open_dynamic_archive`
- `ldemul_after_parse`
- `lang_process()` in `ldlang.c`
 - create `output_bfd`
 - `ldemul_set_output_arch`

- `ldemul_create_output_section_statements`
- read objects, create input bfd's - all symbols exist, but have no values
- may call `ldemul_unrecognized_file`
- will call `ldemul_recognized_file`
- `ldemul_after_open`
- map input sections to output sections
- may call `ldemul_place_orphan` for remaining sections
- `ldemul_before_allocation`
- gives input sections offsets into output sections, places output sections
- `ldemul_after_allocation` - section addresses valid
- assigns values to symbols
- `ldemul_finish` - symbol values valid
- output bfd is written to disk

4 Some Architecture Specific Notes

This is the place for notes on the behavior of `ld` on specific platforms. Currently, only Intel x86 is documented (and of that, only the auto-import behavior for DLLs).

4.1 Intel x86

`ld` can create DLLs that operate with various runtimes available on a common x86 operating system. These runtimes include native (using the mingw "platform"), cygwin, and pw.

auto-import from DLLs

1. With this feature on, DLL clients can import variables from DLL without any concern from their side (for example, without any source code modifications). Auto-import can be enabled using the `--enable-auto-import` flag, or disabled via the `--disable-auto-import` flag. Auto-import is disabled by default.
2. This is done completely in bounds of the PE specification (to be fair, there's a minor violation of the spec at one point, but in practice auto-import works on all known variants of that common x86 operating system) So, the resulting DLL can be used with any other PE compiler/linker.
3. Auto-import is fully compatible with standard import method, in which variables are decorated using attribute modifiers. Libraries of either type may be mixed together.
4. Overhead (space): 8 bytes per imported symbol, plus 20 for each reference to it; Overhead (load time): negligible; Overhead (virtual/physical memory): should be less than effect of DLL relocation.

Motivation

The obvious and only way to get rid of dllimport insanity is to make client access variable directly in the DLL, bypassing the extra dereference imposed by ordinary DLL runtime linking. I.e., whenever client contains something like

```
mov dll_var,%eax,
```

address of `dll_var` in the command should be relocated to point into loaded DLL. The aim is to make OS loader do so, and then make `ld` help with that. Import section of PE made following way: there's a vector of structures each describing imports from particular DLL. Each such structure points to two other parallel vectors: one holding imported names, and one which will hold address of corresponding imported name. So, the solution is de-vectorize these structures, making import locations be sparse and pointing directly into code.

Implementation

For each reference of data symbol to be imported from DLL (to set of which belong symbols with name `<sym>`, if `__imp-<sym>` is found in `implib`), the import fixup entry is generated. That entry is of type `IMAGE_IMPORT_DESCRIPTOR` and stored in `.idata$3` subsection. Each fixup entry contains pointer to symbol's address within `.text` section (marked with `__fuN-<sym>` symbol, where `N` is integer), pointer to DLL name (so, DLL name is referenced by multiple entries), and pointer to symbol name thunk. Symbol name thunk is singleton vector (`__nm.th-<symbol>`) pointing to `IMAGE_IMPORT_BY_NAME` structure (`__nm-<symbol>`) directly containing imported name. Here comes that "on the edge" problem mentioned above: PE specification rambles that name vector (`OriginalFirstThunk`) should run in parallel with addresses vector (`FirstThunk`), i.e. that they should have same number of elements and terminated with zero. We violate this, since `FirstThunk` points directly into machine code. But in practice, OS loader implemented the sane way: it goes thru `OriginalFirstThunk` and puts addresses to `FirstThunk`, not something else. It once again should be noted that `dll` and symbol name structures are reused across fixup entries and should be there anyway to support standard import stuff, so sustained overhead is 20 bytes per reference. Other question is whether having several `IMAGE_IMPORT_DESCRIPTOR`s for the same DLL is possible. Answer is yes, it is done even by native compiler/linker (`libth32`'s functions are in fact resident in windows9x `kernel32.dll`, so if you use it, you have two `IMAGE_IMPORT_DESCRIPTOR`s for `kernel32.dll`). Yet other question is whether referencing the same PE structures several times is valid. The answer is why not, prohibiting that (detecting violation) would require more work on behalf of loader than not doing it.

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